# *Paleographia Digitalis*

## Automated Latin Manuscript Classification

Project report for
Machine Learning for Humanities

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## Introduction

*Paleographia Digitalis* is an academic project that addresses the challenge of automatic manuscript categorization with traditional archival scholarship. It's an application developed using an ensemble of deep learning architecture combining a tailored version of ConvNeXt-Small and EfficientNet-V2-S. It is intended to automate the classification task of historical handwriting in medieval Latin manuscripts into twelve distinct paleographical traditions. The model learns subtle ductus, pen stroke angles, ligature morphologies, and minim structures to classify Latin manuscripts. Ultimately, the project aims at providing a preliminary classification for archival materials, thereby supporting archivists and paleographers in their research.

## Dataset

The model was trained on the ICDAR CLaMM (Classification of Latin Medieval Manuscripts) dataset, specifically: *ICDAR_CLaMM_training*, consisting of 4500 images used in the ICFHR 2017 Competition on the Classification of Medieval Handwritings in Latin Script[1]. The images comprise all types of formats, (JPG, PNG, TIFF), color and greyscale images, while representing twelve historical handwriting categories. The metadata file considered also date classification, nevertheless for this project was considered only the Script label.

## Disclaimer

However, applying automated classification to historical manuscripts deals with a paleographical challenge: the absence of a universally unified nomenclature, both internationally and across national scholarly traditions. Disagreements among paleographical schools regarding the naming and boundaries of specific scripts, particularly late-medieval hands characterized by morphological mixing (*mescidanza*).
Despite these issues, a computational classification attempt remains worthwhile undertaking by prioritizing highly formalized book scripts, while recognizing that transitional or hybrid categories naturally exhibit overlapping features that challenge both human experts and machine learning models alike.
Therefore, the twelve scripts types the CLaMM dataset provides tend to follow reliable formalized scripts that are widely accepted by the paleographical community. (See Image 1)

[1] Cloppet et al., «ICDAR 2017 Competition», Kyoto, 2017

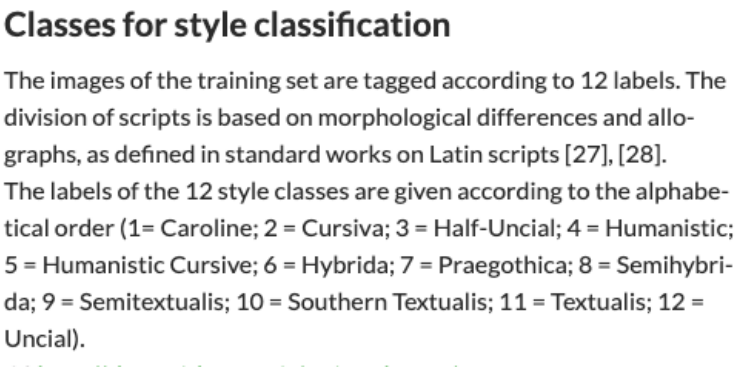



*Image 1: Script types*

## Pre-processing data

First, we pre-process the dataset for the training, to standardize everything and not have errors during the training. The CLaMM dataset (comprising 4,540 high-resolution medieval manuscript images across 12 script types) went through data ingestion, mapping, and a preprocessing pipeline.

The metadata file *CLaMM_Metadata.csv* was parsed to convert the categorical SCRIPT_TYPE attribute into zero-indexed integer targets (*0* to *11*), while all image files were indexed into an absolute path dictionary (image_map) to guarantee constant-time $O(1)$ file retrieval regardless of directory structures or extension variations. Then we disable Pillow's pixel limits (MAX_IMAGE_PIXELS = None) to prevent errors and memory time when handling large-scale archival scans.

To manage data during training, a custom class (*CLaMMDataset*) was created to deal with case-insensitive filename indexing and automated extension fallback routines (.jpg, .jpeg, .tiff, .png). Alongside the images were standardized into the RGB color space and processed through a transformation pipeline that resized manuscript pages to 224x224 pixels, converted to tensors, and normalized them using standard ImageNet parameters.

In this pre-processing phase the 4500 images were partitioned using an 80/20 train-validation split with a fixed pseudo-random seed (seed = 42), as a result 3632 training images and 908 validation images.

Finally, data loaders were configured with a batch size of 32 and multi-process parallel reading (num_workers = 2) to maximize GPU utilization and prevent CPU-RAM input-output bottlenecks during model execution.

## The Model

Following the initial setup, the model was built upon a standard ResNet-18 architecture pre-trained on ImageNet at 224x224 resolution, with Adam optimizer ($learning_rate = 10^{-4}$)over 10 epochs. The results were a high training accuracy (99.34%) while validation accuracy reached a 63,55%, this demonstrated a severe overfitting during the training, meaning the network memorized and it did not learn stroke features.

Following this first training loop, we decided to try and use a new architecture to address the spatial overfitting, now using ConvNeXt-Tiny at 224x224 resolution. In this new loop a more appropriate domain data augmentation pipeline was implemented, with : RandomResizedCrop, controlled rotations ($\pm 8^{\circ}$), and ColorJitter to simulate parchment aging and ink degradation, new optimization approach with AdamW ($weight_decay = 10^{-2}$) paired with Label Smoothing Loss ($0.1$). This setup improved the validation accuracy bringing it to 71.59% ($MacroF_1 = 0.7224$), yet not the result we wanted to achieve.

To improve our model, we considered the critical paleographical markers that make the model learn, we realize that the specific details of the scripts were lost in the low resolution we were using. Therefore, we implemented another new architecture EfficientNet-V2-S, utilizing native SiLU (Swish) activations that capture the ductus, pen stroke angles, ligatures, and minim structures of the scriptures as well the resolution was scaled to 384x384 pixels. Furthermore, to optimize convergence during this advanced training phase, we incorporated Class-Weighted CrossEntropyLoss, applied a higher penalty weight to difficult classes (Types 6, 8, and 9), forcing the model to learn and preventing it from ignoring it. It was also added an Extended Fine-Tuning & Cosine Annealing: over 20 epochs to keep the learning rate low and controlled, preventing the abrupt destruction of pre-trained feature representations. Finally, we integrated domain-aware augmentations, RandomResizedCrop, ColorJitter and controlled RandomRotation to enhance without altering authentic script morphology.
The results of this training loop were one of the best outcomes we had, since the Validation accuracy increased to 78.19% ($ValidationLoss \approx 1.05$).

However, we tried to obtain even higher results, attempting several simulations changing parameters to see if we were able to increase the accuracy, nevertheless these attempts did not increase, instead they stalled under the accuracy that we had with EfficientNet-V2-S.
The alternative strategies were:

- Tiling and Patch Cropping: Sub-sampling page patches caused spatial fragmentation, constraining validation accuracy between 72.25% and 74.12%.
- Test-Time Augmentation (TTA): Applying horizontal flips during inference reduced accuracy to 76.76%. In paleography the stroke ductus and script inclination carry inherent directionality; mirror flipping alters these features and introduces classification noise.

One last attempt was performed to improve the accuracy; we address the limits of EfficientNet-V2-S now changing into ConvNeXt-Small at 384x384 resolution, trained with a Cosine Annealing learning rate schedule across 20 epochs. The new architecture has larger 7x7 depthwise kernels, effectively capturing macroscopic paragraph structure and line continuity alongside local character strokes. This ultimate training loop reach validation accuracy of 79.74% (peak at epoch 14) with improved loss stability ($ValidationLoss \approx 0.87 - 0.94$). The best output we obtain after several attempts.

The final strategy to exceed the 80% accuracy threshold, was a weighted soft-voting probability ensemble to combine the complementary feature representations of ConvNeXt-Small and EfficientNet-V2-S. It combined complementary strengths of the two leading models using Softmax Probability Averaging:

$$P_{Ensemble} = 0.60 \cdot P_{ConvNeXt-Small} + 0.40 \cdot P_{EfficientNet-V2-S}$$

The outcome of the ensemble model achieved the highest and top performance of our learning model with 80.40% Validation Accuracy.

## Evaluation

The performance of the final ensemble model (60/40) was evaluated across all $N = 908$ validation images, generating the following classification metrics.

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.90 | 0.93 | 0.91 | 120 |
| 1 | 0.78 | 0.75 | 0.76 | 110 |
| 2 | 0.97 | 0.88 | 0.92 | 41 |
| 3 | 0.85 | 0.96 | 0.90 | 53 |
| 4 | 0.93 | 0.93 | 0.93 | 44 |
| 5 | 0.70 | 0.68 | 0.69 | 79 |
| 6 | 0.86 | 0.86 | 0.86 | 107 |
| 7 | 0.66 | 0.56 | 0.61 | 59 |
| 8 | 0.63 | 0.57 | 0.60 | 70 |
| 9 | 0.78 | 0.78 | 0.78 | 40 |
| 10 | 0.75 | 0.82 | 0.78 | 147 |
| 11 | 1.00 | 0.97 | 0.99 | 38 |
| accuracy | | | 0.80 | 908 |
| macro avg | 0.82 | 0.81 | 0.81 | 908 |
| weighted avg | 0.80 | 0.80 | 0.80 | 908 |

*Image 2. Classification Report*

The report illustrates overall the strong and weaker strengths of the model considering the difficulties of the script's characteristics. For example:

- Uncial (Class 11) demonstrated optimal separability ($Precision = 1.00, F_1 = 0.99$)
- Caroline (Class 0), Half-Uncial (Class 2), Humanistic (Class 3), and Humanistic Cursive (Class 4) maintained high $F_1$ scores ($\geq 0.90$), indicating well-defined letterforms successfully captured
- Semihybrida (Class 7) and Semitextualis (Class 8) recorded lower performance ($F_1 = 0.61$ and $0.60$). This represents paleographical challenges such as the transition from Gothic scripts that share significant visual and structural overlap.

Alongside the report we created a confusion matrix, which provides a detailed overview of the classification accuracy across all 12 categories.

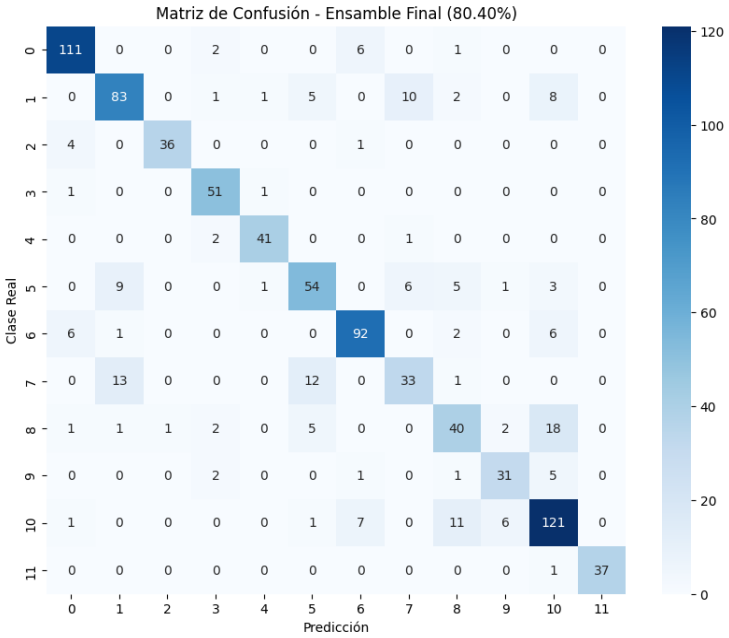



*Image 3. Confusion Matrix*

The matrix demonstrates a high concentration of numbers along the primary diagonal that represents the model reliably identifying the distinct paleographical patterns and ductus structures.

Overall, the presence of minor residual misclassifications remain visible primarily between closely related script types sharing overlapping historical or morphological traits (such as Classes 1, 5, 7, and 8), as expected given the subtle variations in some styles.

Ultimately the Macro $F_1$ metric (*0.81*) exceeding the Weighted $F_1$ metric (*0.80*) confirms that the ensemble maintains balanced performance across minority and majority classes, rather than relying solely on overrepresented script categories.

## Final Application

The trained model was integrated into a functional web application named *Paleographia Digitalis*. Developed using the Flask python library, the application provides an intuitive interface that allows users to upload manuscript images for automated classification and prediction. Alongside the uploaded image, a percentage table displays the results, with the primary prediction identifying the historical script. Beneath this it appears a *Paleographical Dossier* that provides a comprehensive overview of the script type, including its common and Latin names, geographical region, active time period, alongside its general characteristics and historical examples of manuscripts. Additionally, the application features a catalogue of the twelve supported scripts, allowing users to browse and view the detailed dossier for each individual historical writing style.

At last, all the work produced during this project, including the Jupyter notebooks for the model creation, the dataset, the application files and requirements, is publicly available in the GitHub repository, alongside a live link to the online interface of *Paleographia Digitalis.*

## References


- Bischoff, B. (2009). Paläographie des römischen Altertums und des abendländischen Mittelalters (4th ed.). E. Schmidt.

- Clark, A. (2015). Pillow (PIL Fork) Documentation. https://pillow.readthedocs.io/

- Cloppet, F., Eglin, V., Helias-Baron, M., Kieu, C., Stutzmann, D., & Vincent, N. (2017). ICDAR 2017 competition on the classification of medieval handwritings in Latin script (CLaMM). 2017 14th IAPR International Conference on Document Analysis and Recognition (ICDAR), Vol. 1, pp. 1371–1376. Kyoto, 2017. https://doi.org/10.1109/ICDAR.2017.224

- Derolez, A. (2003). The Palaeography of Gothic Manuscript Books from the Twelfth to the Early Sixteenth Century. Cambridge University Press.

- Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 11976–11986.

- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems (NeurIPS), 32, 8026–8037.

- Stutzmann, D., & Helias-Baron, M. (2017). ICDAR 2017 Competition on the Classification of Medieval Handwritings in Latin Script - Dataset [Data set]. Zenodo. https://doi.org/10.5281/zenodo.5527690

- Tan, M., & Le, Q. V. (2021). EfficientNetV2: Smaller models and faster training. International Conference on Machine Learning (ICML), PMLR, pp. 10096–10106.

- The Pallets Projects. Flask Documentation. https://flask.palletsprojects.com/